

Problem 1: Jowulu

(17 marks)

Here are some arithmetic equations in Jowulu¹. All numbers are less than 200.

$$\begin{aligned}\text{fulpun} \times \text{fulpun} &= \text{yaa b3ei vapfire} \\ \text{tempun} \times \text{fuuli} &= \text{bvunfulpun} \\ \text{bvunfulpun} + \text{b3ei} &= \text{k\text{on} vate\text{e}} \\ \text{b3ei} \times \text{fuuli} &= \text{tont\text{e}} \\ \text{pfire} + \text{pfire} &= \text{fulpun} \\ \text{fuuli}^{\text{te\text{ena}}} &= \text{fuuli} \\ \text{bvunt\text{e}} \times \text{tempun} &= \text{yaa pfire vabvunt\text{tempun}} \\ \text{ta\text{a}}^{\text{b3ei}} &= \text{yaa tont\text{e} vata\text{a}}\end{aligned}$$

Assignment 1: (8 marks)

Rewrite the equations above with regular (Hindu-Arabic) numerals (like “1 + 1 = 2”).

Assignment 2: (6 marks)

Write in Jowulu: (a) 78; (b) 185; (c) 25.

Assignment 3: (3 marks)

Write with regular (Hindu-Arabic) numerals:

(a) yaa tont\text{e} vabvunt\text{tempun}

(b) yaa pfire vata\text{a}